

GRODNO, Belarus

WMO: 268250

Lat: 53.602N

Lon: 24.056E

Elev: 490

StdP: 14.44

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	3.8	-7.6	3.7	-2.0	-1.0	5.3	4.3	28.1	39.6	25.1	35.5	4.2	350	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	85.0	68.0	81.2	66.3	78.3	64.6	70.0	80.4	68.2	78.2	66.5	75.4	8.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	99.4	75.0	64.6	93.0	72.4	62.9	87.4	70.6	34.2	80.7	32.8	77.9	31.4	75.7	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.6	19.8	17.1	-9.1	90.2	6.8	3.7	-14.0	92.8	-18.0	95.0	-21.8	97.1	-26.7	99.8		
			-9.5	73.1	6.7	2.4	-14.3	74.9	-18.2	76.3	-22.0	77.6	-26.9	79.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	25.4	27.5	34.8	46.7	55.7	61.0	66.1	64.6	55.5	45.2	36.6	28.9	
	DBStd	16.25	10.74	10.08	8.08	7.84	7.26	5.82	5.75	5.82	5.97	7.14	7.78	9.81	
	HDD50	3300	763	629	473	153	26	1	0	0	19	181	402	653	
	HDD65	7216	1228	1049	937	550	299	148	56	79	288	613	851	1118	
	CDD50	1759	0	0	1	54	203	331	498	453	185	33	1	0	
	CDD65	200	0	0	0	1	11	28	90	66	4	0	0	0	
	CDH74	1808	0	0	0	15	122	246	807	579	39	0	0	0	
	CDH80	454	0	0	0	1	14	37	236	160	6	0	0	0	
Wind	WSAvg	7.8	9.1	8.8	8.8	8.0	7.4	6.8	6.5	6.5	7.2	7.6	8.5	8.9	
Precipitation	PrecAvg	22.9	1.4	1.3	1.3	1.4	2.3	2.5	3.4	2.3	2.0	1.7	1.7	1.6	
	PrecMax	27.6	3.1	2.4	2.9	3.0	4.5	4.9	6.8	6.4	5.4	5.0	3.0	2.6	
	PrecMin	17.4	0.4	0.4	0.4	0.4	0.9	0.7	0.2	0.2	0.6	0.1	0.5	0.6	
	PrecStd	2.6	0.6	0.5	0.6	0.7	0.8	1.1	1.7	1.3	1.4	1.2	0.6	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.2	47.6	59.2	75.7	82.1	84.3	90.9	90.9	78.7	67.0	54.8	48.1	
		MCWB	43.9	43.2	48.8	58.2	64.5	68.0	69.0	68.6	62.4	59.2	50.9	45.6	
	2%	DB	42.1	44.2	54.0	69.4	77.4	80.3	86.2	84.3	73.3	61.8	51.4	44.7	
		MCWB	40.3	41.5	46.2	55.2	63.2	65.6	68.0	68.5	61.0	56.2	49.2	43.2	
	5%	DB	39.4	41.2	49.3	64.5	73.5	76.8	82.2	80.2	69.4	58.4	48.4	42.2	
		MCWB	37.8	39.0	43.5	52.0	60.4	64.0	67.3	65.7	59.7	54.0	46.6	40.9	
	10%	DB	37.1	38.8	45.1	60.3	69.4	73.2	78.5	76.6	66.0	55.4	46.4	39.5	
		MCWB	35.9	37.1	41.2	49.9	58.4	62.0	65.6	63.6	57.7	52.2	44.9	38.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.0	44.3	50.8	60.7	67.4	70.6	73.4	72.3	65.5	60.7	52.2	46.3
			MCDB	45.4	46.4	56.8	74.9	78.1	81.2	85.3	82.7	74.7	66.1	54.4	47.3
2%		WB	40.5	41.5	46.7	56.2	64.6	67.8	70.4	69.5	62.5	57.3	49.4	43.3	
		MCDB	41.6	43.4	52.3	66.6	74.5	77.1	80.6	80.5	69.9	60.6	51.1	44.3	
5%		WB	37.8	39.2	43.9	53.7	62.2	65.6	68.7	67.5	60.6	54.7	46.8	40.7	
		MCDB	38.9	40.8	48.1	62.8	71.0	74.1	78.6	77.5	66.8	57.4	48.1	42.0	
10%		WB	35.8	37.1	41.5	51.0	59.6	63.3	66.9	65.5	58.8	52.5	44.9	38.3	
		MCDB	36.8	38.7	44.8	58.2	67.4	70.8	76.0	73.5	64.6	55.2	46.1	39.3	
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.6	11.9	17.0	18.5	17.6	18.3	18.4	15.8	11.8	7.1	7.0	
		MCDBR	7.1	9.5	18.4	24.7	24.4	23.1	24.2	24.1	21.7	16.3	9.6	7.5	
		MCWBR	6.8	7.7	11.9	12.7	12.1	11.5	10.1	9.8	11.1	10.8	7.8	7.0	
	5% WB	MCDBR	6.7	9.0	16.2	22.3	21.5	20.2	20.7	20.5	18.7	14.3	8.8	7.3	
		MCWBR	6.6	7.7	11.3	12.1	11.8	11.0	10.0	9.4	10.6	10.4	7.6	7.1	
Clear-Sky Solar Irradiance	taub	0.299	0.332	0.366	0.413	0.400	0.391	0.428	0.422	0.382	0.357	0.341	0.300		
	taud	2.207	2.181	2.220	2.185	2.289	2.338	2.251	2.264	2.341	2.397	2.292	2.244		
	Ebn at Noon	206	237	255	258	269	272	259	253	248	229	189	182		
	Edh at Noon	20	29	35	41	39	37	40	38	31	24	19	16		
All-Sky Solar Radiation	RadAvg	208	446	846	1294	1669	1757	1637	1414	979	505	188	129		
	RadStd	36	81	133	167	167	147	166	128	125	103	20	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (14 neighbors)	N/A	N/A	N/A	N/A	+1.12	N/A	N/A	N/A	+225	+74
	N/A	N/A	N/A	N/A	+0.83	+1.10	N/A	N/A	N/A	N/A

Nomenclature: See separate page